IV SEMESTER	L	T	P	C
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20CS4L02 : OPERATING SYSTEMS USING LINUX LAB				

COURSE OUTCOMES:

At the end of the course students are able to

CO1: Interpret with the UNIX/LINUX environment

CO2: Infer the fundamentals of shell scripting/programming

CO3: Describe CPU scheduling and Write programs on it.

CO4: Demonstrate Bankers Algorithm for Dead Lock Avoidance.

CO5: Explain Disk allocation methods and Write programs on it.

CO6: Describe Page Replacements algorithms and Write programs on it.

LIST OF PROGRAMS:

1. a) Study of Vi editor, Bash Shell, Bourne Shell and C Shell in UNIX/LINUX Operating System.

b) Study of UNIX/LINUX general purpose utility command list man,who,cat, cd, cp, ps, ls, mv,rm,mkdir, rmdir, echo, more, date, time, kill, history, chmod, chown, finger, pwd, cal, logout, shutdown.

c) Study of Unix/Linux file system (tree structure).

e) Study of .bashrc, /etc/bashrc and Environment variables.

2. UNIX/LINUX based exercises to practice/simulate File system related system calls and some of the process management concepts in LINUX Environment.

a) Write a C program that makes a copy of a file using standard I/O, and system calls

b) Write a C program to emulate the UNIX ls –l command.

c) Write a C program that illustrates how to execute two commands concurrently with a command pipe Ex: - ls –l | sort.

d) Write a C program that illustrates two processes communicating using shared memory.

e) Write a C program to simulate producer and consumer problem using semaphores.

f) Write C program to create a thread using pthreads library and let it run its function.

g) Write a C program to illustrate concurrent execution of threads using pthreads library.

3. Simulate FCFS and SJF CPU scheduling algorithm.
4. Simulate Priority CPU scheduling algorithm.
5. Simulate Round Robin CPU scheduling algorithm.
6. Simulate Bankers Algorithm for Dead Lock Avoidance.
7. Simulate Sequential file allocation strategy.
8. Simulate Linked file allocation strategy.
9. Simulate Indexed file allocation strategy.
10. Simulate First In First Out page replacement algorithm.
11. Simulate Least Recently used page replacement algorithm.
12. Simulate optimal page replacement algorithm.

Text Books:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, "Operating System Concepts", 10th Edition, John Wiley and Sons Inc., 2018.
2. William Stallings, "Operating Systems– Internals and Design", 7th Edition, Prentice Hall, 2016.
3. Alex A Aravind, Operating Systems-S Halder, Second Edition, Pearson Education, 2016.
4. Andrew Tanenbaum, Herbert Bos, "Operating Systems", 4th Edition, 2015.

Reference Books:

1. Ann McIver McHoes Ida M. Flynn, "Understanding Operating Systems" Sixth Edition, Course Technology-Cengage Learning, 2011.
2. Andrew S. Tanenbaum, "Modern Operating Systems", Second Edition, Addison Wesley, 2001.
3. Andrew S. Tanenbaum, Albert S. Woodhull - Amherst, "Operating Systems Design and Implementation", Third Edition, Prentice Hall, 2006.
4. W. Richard Stevens, "Advanced Programming in UNIX Environment", 2nd Ed, Pearson Education, 2005.
6. Terrence Chan, "UNIX System Programming Using C++", Prentice Hall India, 1999.

E-Reference:

http://nptel.iitm.ac.in/courses/Webcourse-contents/IIScBANG/Operating%20Systems/New_index1.html